

Question Paper

Exam Date & Time: 30-Dec-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU

THIRD SEMESTER B.TECH. DEGREE MAKEUP EXAMINATIONS - DECEMBER 2025

MATHEMATICAL FOUNDATIONS FOR DATA SCIENCE-I [MAT_2127]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) The equation of two regression lines obtained in a correlation analysis is as follows: $3x + 2y = 19$, (4)
 $3y + 9x = 46$

- A) (I) Obtain mean of x and y
(II) Obtain the value of correlation coefficient

Course Outcome: "CO1" / Cognitive Level: "Application"

- B) Fit the equation of the form $y = ax + b$ to the following data: (3)

x	1	2	4	6	7	8
y	2.4	3	3.6	4	5	6

Course Outcome: "CO1" / Cognitive Level: "Application"

- C) It is suspected that a patient has one of the diseases A_1, A_2, A_3 . Suppose that the population (3)
suffering from this illness are in the ratio 2:1:1. The patient is given a test which turns out to be
positive in 25% of the cases of A_1 , 50% of the cases of A_2 and 90% of the cases of A_3 . Given that
out of 3 tests taken by the patient two are positive, then find the probability for each of the diseases.

Course Outcome: "CO1" / Cognitive Level: "Application"

- 2) The random variable (X, Y) has a joint pdf given by (4)

- A) $f(x, y) = x + y$, $0 \leq x \leq 1$, $0 \leq y \leq 1$. Compute the correlation (4)
coefficient between X and Y .

Course Outcome: "CO2" / Cognitive Level: "Application"

- B) Suppose that the joint pdf of (X, Y) is given by (3)

$$\text{If } f(x, y) = \begin{cases} \frac{2}{a^2} & 0 \leq x \leq y \leq a \\ \text{elsewhere} & \end{cases}.$$

Find the conditional pdf of $Y = y$ for given $X = x$.

Course Outcome: "CO2" / Cognitive Level: "Application"

- C) (3)

Let X be a continuous random variable with probability density function $f(x)$ is defined by:

$$f(x) = \begin{cases} kx^4, & 0 < x < 1 \\ 0, & \text{elsewhere} \end{cases}$$

Find

- (i) The value of k .
- (ii) $P\left(\frac{1}{4} < X < \frac{3}{4}\right)$.

Course Outcome: "CO2" / Cognitive Level: "Application"

- 3) The times to failure in months X of the light bulbs produced at two manufacturing plants A and B of a company obey exponential distributions with means 5 and 2 months respectively. Plant B produces three times as many bulbs as Plant A. The company mixes the produced bulbs at both the plants before selling. What is the probability that a randomly selected bulb from the mixed lot will work for at least 5 months?
- A)

Course Outcome: "CO3" / Cognitive Level: "Application"

- B) The number of telephone lines busy at an instant of time is a binomial variate with a probability of 0.2. If at an instant 10 lines are chosen at random, what is the probability that
- (I) No line is busy
 - (II) At least one line is busy
 - (III) At most two lines are busy

Course Outcome: "CO3" / Cognitive Level: "Application"

- C) Derive an expression for expectation of a continuous random variable having Chi-square distribution with 'n' degree of freedom $(\chi^2_{(n)})$.

Course Outcome: "CO3" / Cognitive Level: "Application"

- 4) If X and Y are independent random variables and have exponential distributions with parameter 1, find the pdf of $U = X - Y$.
- A)

Course Outcome: "CO4" / Cognitive Level: "Application"

- B) Let $X_1, X_2, \dots, X_n$ denote a random sample of size n from a distribution having pdf
- $$f(x, \theta) = \begin{cases} \theta^x (1 - \theta)^{1-x}, & 0 \leq \theta \leq 1 \\ 0, & \text{elsewhere} \end{cases}$$
- Find the maximum likelihood estimation for θ .

Course Outcome: "CO4" / Cognitive Level: "Application"

- C) If $\bar{X}$ is the mean of a random sample from a normal distribution with mean μ and variance 100, find n such that
- $$P(\mu - 5 < \bar{X} < \mu + 5) = 0.954.$$

Course Outcome: "CO4" / Cognitive Level: "Application"

- 5) Following are the lifetime of four batches of electric lamps. Perform an analysis of variance of this data and test the homogeneity of four batches. (5)

A)

1	1600	1610	1650	1700	1720	1800	1680	
2	1580	1660	1640	1700	1750			
3	1460	1550	1600	1640	1620	1660	1740	1820
4	1510	1520	1530	1570	1600	1680		

Course Outcome: "CO5" / Cognitive Level: "Application"

- B) Fit a Poisson distribution to the following data and test the goodness of fit. (5)

X	0	1	2	3	4	5	6
f	275	72	30	7	5	2	1

Course Outcome: "CO5" / Cognitive Level: "Application"

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Event Name	Subject
058-MIT BLR-MAKEUP EXAM-2025-2026	MAT_2127:MATHEMATICAL FOUNDATIONS FOR DATA SCIENCE-I

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
058-MIT BLR-MAKEUP EXAM-2025-2026	MAT_2127:MATHEMATICAL FOUNDATIONS FOR DATA SCIENCE-I

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	14	50(100%)
4	Analysis	0	0(0%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		14	50 (100%)